Lesson notes · for you

Build an agent in Python: unit check

AI agents · Upper secondary · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Read a tool call
- Tools are functions
- The agent loop
- Guardrails in code

Activity: Activity: Build the agent on paper

1. Warm-up (3 min): Without notes, write the four parts: the parser, the tools, the loop and the guard.
2. Draw (7 min): Draw the whole agent: the model's text going in, the parser, the guard, a person, the tools and the history, with arrows.
3. Break it (7 min): Swap drawings. Your partner invents a model output that gets through, such as a broken call, a made-up tool, a model that never answers or a message nobody approved.
4. Fix (3 min): Add the rule, or the test, that would have stopped it.

Use made-up tools, places and people only.

Discussion

- Which part of your agent was hardest to get right?
- What could a real model do that your scripted model never did?
- Where would you put a person in the loop, and why?

Check yourself

- One guardrail I would add to every agent, and why.

Try it this week

Pick one part of your agent (the parser, the tools, the loop or the guard) and add one test that would have caught a real mistake from this unit, such as 9Z copied as 9C or a guard checking the wrong tool name.

Worksheet

Build an agent in Python: unit check

AI agents · Upper secondary · Unit check

Name _____ Class _____ Date _____

1 **On each turn, what does an agent's model actually produce?**

- ◯ Text: a plan, a tool call or an answer
- ◯ The tool's result
- ◯ A finished action, such as a sent message

2 **json.loads turns the JSON text of a call's details into a Python dictionary.**

- ◯ True
- ◯ False

3 **Put the lines of run in order.**

Number the lines in order (1, 2, 3...). Cross out any line that doesn't belong.

- ☐ `return {"error": "unknown tool"}`
- ☐ `def run(name, args):`
- ☐ `return TOOLS[args](name)`
- ☐ `if name not in TOOLS:`
- ☐ `return TOOLS[name](args)`

4 **What does this program print?**

```
import json
text = 'weather{"place":"Kafi"'
try:
    json.loads(text[text.find("{"):])
    print("call")
except ValueError:
    print("broken call")
```

- ◯ call
- ◯ broken call
- ◯ Nothing: an error stops the program

5 **Match each part of your agent to what it protects against.**

try / except ValueError	• • • •	A message nobody approved
if name not in TOOLS	• • • •	A model that never answers
for step in range(limit)	• • • •	A made-up tool causing a KeyError
approved != args	• • • •	A broken call crashing the agent